Operation explanation of AILEIKE portable air pump ALKB602, pressure gauge calibration is no longer a hassle

The AILEIKEt ALKB602 is a lightweight manual pneumatic pressure generating device. Its pressure-generating medium is clean air, capable of producing both positive and negative (vacuum) pressure outputs. The device features an open and portable structure, boasting a lightweight design and stable pressure regulation. Equipped with dual output interfaces, it can simultaneously connect to a standard gauge and the instrument being verified, accommodating both laboratory metrological verification and industrial on-site instrument calibration tasks. It is widely used for pressure generation and calibration of medium and low-pressure pneumatic instruments.

1. Technical indicators

name	Portable air pressure pump	Pressurization method	manual
model	ALKB602	medium	clean air
pressure range	-0.095～6MPa	Overall dimensions	310×250×180mm
Adjust fineness	10 Pa	weight	4.2kg
Output port	2 pieces	Pressure connection	Quick-connect thread (M20*1.5)

II. Operation method

First, place the lightweight air pressure pump on a flat workbench (in a non-corrosive, vibration-free, and dust-free environment).

Install the standard gauge and the verified pressure instrument (pressure transmitter, pressure gauge, pressure sensor, etc.).

Positive pressure calibration process

1. Voltage boosting process

First, open the vent valve of ALKB602, then push the positive and negative pressure switch valve inwards (do not switch under pressure), and finally, turn the fine adjustment knob counterclockwise until it stops, and close the vent valve.

Use the hand pressure lever to apply pressure, stop when the pressure reaches the verification point, and then use the fine adjustment to slowly turn clockwise to reach the verification point.

2. Decompression process

First, slowly open the vent valve and reduce the pressure until it stops at the verification point. Then, close the vent valve and use the fine adjustment valve to slowly adjust it counterclockwise to the verification point.

3. Pressure relief process

Open the vent valve, gradually release the pressure in the pipeline, and when the pressure is released to 0, the measurement is complete.

Vacuum calibration process

1. Voltage boosting process

Open the vent valve, then pull out the positive and negative pressure switch valve (do not switch under pressure), adjust the ALKB602 fine adjustment knob clockwise until it stops, and close the vent valve.

Use the hand-operated pressure rod to evacuate the vacuum until it reaches approximately the verification value, and then use the fine adjustment to slowly turn counterclockwise to reach the verification point.

2. Decompression process

First, slowly open the vent valve to reduce the pressure to around the verification value and then stop. Close the vent valve and use the fine adjustment valve to slowly adjust clockwise to the verification point.

3. Pressure relief process

Open the vent valve, gradually release the pressure inside the pipeline until it reaches 0, and then the measurement is complete.

III. Application Scenarios

1. Industrial site maintenance: In factory workshops and pipeline sites, quickly calibrate air pressure gauges, pressure switches, and air pressure sensors without requiring complex equipment, and use them immediately upon retrieval.

2. Laboratory calibration: The metrology room and quality inspection laboratory are used for calibrating various small pressure instruments, ensuring sufficient accuracy and convenient operation.

3. Instrument maintenance and debugging: When maintenance personnel overhaul faulty pressure instruments, they use pressure testing to determine whether the instrument is accurate and functioning properly.

4. Outdoor portable operations: Field equipment debugging, construction site instrument testing, and normal operation even in scenarios without power supply or external gas source.

5. Teaching and Training: For teaching related to electromechanical and instrumentation in colleges and universities, used for demonstrating pressure testing training.

IV. Precautions

1. The equipment should be used within the rated pressure range as much as possible, and overrange use is prohibited.

2. It is prohibited to push or pull the positive and negative pressure switching valve under pressure. The positive and negative pressure modes must be switched only after the pressure is completely reduced to zero to avoid internal damage.

3. The calibration bench should be placed on a stable operating platform for use, and should be avoided being used in environments with corrosive gases and dust particles.

4. Only clean air is allowed for pressure generation. It is strictly prohibited to introduce oil, water, or corrosive

gases to avoid contaminating the interior of the pump and causing blockages or leaks.

5. When applying pressure with a lever, ensure even application of force. Do not operate the valve violently when closing or opening it.

6. For long-term storage, clean the interfaces, block the output ports with plugs, store in a dry and ventilated environment, and regularly check the flexibility of the valve components.

V. Conclusion

The ALKB602 portable air pressure pump integrates positive and negative pressure generation capabilities, featuring a lightweight and portable design with stable and reliable pressure output. It can meet the needs of laboratory metrological verification and is also suitable for on-site inspection and calibration in industrial environments. Strict adherence to this specification for operation and maintenance can ensure the accuracy of equipment calibration, extend equipment lifespan, and make it an ideal pressure generating device for the calibration of medium and low pressure air pressure instruments.